Diagnosing Bias and Variance 🩺

When a model doesn't work well on the first try, the key is to determine *why*. The most systematic way to do this is to diagnose whether the model is suffering from **high bias (underfitting)** or **high variance (overfitting)**.

This is done by comparing the error on the **training set (J_{train})** with the error on the **cross-validation set (J_{cv})**.

Identifying the Problem

Here are the key indicators for diagnosing your model:

Diagnosis	Key Indicator(s)	Description
High Bias (Underfitting)	J_{train} **is high** (and $J_{cv} \approx J_{train}$)	The model is too simple. It's not even fitting the training data well.
High Variance (Overfitting)	$J_{cv} \gg J_{train}$ (and J_{train} is low)	The model is too complex. It fits the training data perfectly but fails to generalize to new (CV) data. The *gap* is the key.
"Just Right"	J_{train} **is low** (and $J_{cv} \approx J_{train}$)	The model fits the training data well and *also* generalizes well to new data. Both errors are low.

Visualizing Error vs. Model Complexity

A very powerful way to see this relationship is to plot the training and CV errors as a function of model complexity (like the degree of polynomial, `d`).

- **Horizontal Axis:** Model Complexity (e.g., polynomial degree `d` from 1 to 10).
- **Vertical Axis:** Error.

Behavior of the Error Curves

- **Training Error (J_{train}):**
  - As model complexity (`d`) **increases**, the training error **consistently decreases**.
  - A more complex model (like a 10th-order polynomial) can fit the training data almost perfectly, so J_{train} will be very low.

- **Cross-Validation Error (J_{cv}):**
    - This curve has a characteristic **U-shape**.
    - **At low complexity (e.g., d=1):** J_{cv} is **high** because the model is underfitting (high bias).
    - **At high complexity (e.g., d=10):** J_{cv} is also **high** because the model is overfitting (high variance).
    - **In the middle:** J_{cv} reaches a minimum "sweet spot" (e.g., d=2), which represents the "just right" model.

Diagnosing the Regions of the Graph

- **High Bias Region (Left side):** J_{train} is high, and J_{cv} is high (and close to J_{train}).
- **High Variance Region (Right side):** J_{train} is low, but J_{cv} is **much higher** than J_{train}.
- **"Just Right" Region (Middle):** Both J_{train} and J_{cv} are low and close to each other.

A Special Case: High Bias AND High Variance

While less common (especially in linear models), it is possible for a model (like a neural network) to suffer from *both* problems simultaneously.

- **Indicator:**
    1. J_{train} **is high** (sign of high bias).
    2. J_{cv} **is *even much higher* than** J_{train} (sign of high variance).
- **Intuition:** This can happen if a model is underfitting in some parts of the input space while simultaneously overfitting in other parts.